# Vitaly Chait

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## PROFESSIONAL SUMMARY

Software engineer with experience building production backend systems and hardware diagnostics, integrating data accelerators, evaluating performance against CPU baselines, and deploying complex systems into customer environments. Skilled in Python, C, SQL, and automated testing, with a track record of cross-team integration and customer-site deployment.

## TECHNICAL SKILLS

**Languages:** Python, C, C++, SQL, Java, Bash, PowerShell, JavaScript, TypeScript, MATLAB
**DevOps & Systems:** Linux, Windows, Docker, Git, Jenkins CI/CD, Grafana, Prometheus, BIOS/UEFI, Secure Boot, PerfMon
**Backend & Cloud:** REST APIs, Microservices, Flask, FastAPI, WebSockets, AWS EC2 & S3, Distributed Systems
**Data Infrastructure:** Apache Spark, Presto, ClickHouse, MySQL, Elasticsearch, InfluxDB, scikit-learn, OpenCV
**AI:** RAG, MCP, OpenClaw, LangChain, Unsloth, Hugging Face, AI Harnesses, Claude AI Workflows

## PROFESSIONAL EXPERIENCE

**Software Engineer — NeuroBlade (acquired by AWS)** Jan 2023 – Sep 2025

*Tel Aviv, Israel*

- As part of the acceleration team, integrated data accelerator with Presto/Spark, testing 3TB+ workloads with TPC-H/TPC-DS benchmarks, helping improve execution speed from 0.8× to 4× the CPU-only baseline.
- Identified hardware stalls and kernel throughput bottlenecks; wrote software for embedded hardware kernel components.
- Designed C-based diagnostics across PCIe, NVMe, switches, and FPGAs, including JTAG-based FPGA diagnostics; performed hardware bring-up and utilization testing, monitored system power draw using Schneider Electric UPS telemetry, and created automated workflows for firmware upgrades with cold reboots, reducing bottlenecks from months to weeks.
- Built robust Jenkins CI/CD and Debian release pipelines; configured Kubernetes clusters and Grafana, Prometheus, MySQL, and InfluxDB monitoring; used AI to generate test scripts, analyze logs, and summarize benchmark results.
- Led cross-team integration, increasing automated test count by >1,000% and cutting integration time by over 50%.
- Supported data-accelerator integration and deployment at customer sites with customer engineering teams.

**Software Engineer — Self-employed / Independent Contractor** Jul 2021 – Jan 2023

*Beijing, China*

- Designed and deployed production backend services and REST APIs for SoC-based products using Flask, Python, and SQL; applied TFLite supervised ML models with OpenCV for image and motion detection.
- Helped create low-cost software-hardware interfaces for embedded products.

**System Engineer — Orbotech (Acquired by KLA)** Aug 2018 – Jul 2021

*Tel Aviv, Israel*

- Led system engineering for Micro-LED deployments across sensor fusion, optics, motion algorithms, and software; enabled detection of defects with lateral size <0.5 micron and height <0.1 micron at 60% gross margin.
- Researched optical imaging systems, prototyping in Python and MATLAB with engineering teams and customers.
- Initiated proof-of-concept (POC) projects and facilitated multidisciplinary brainstorming sessions with technical staff and PMs to evaluate solutions.
- Prepared and presented executive reviews to VPs, communicating POC findings, technical risks, and recommendations to guide development priorities.
- Provided 3rd-tier escalation support and led customer success meetings to resolve Micro-LED deployment issues.
- Outstanding Employee, FPD R&D (Jun 2019) — selected from 200+ employees.

**Resident Engineer — Orbotech (Acquired by KLA)** Aug 2017 – Jul 2018

*Beijing, China*

- Customer success liaison between international R&D and field engineering teams during customer deployments.

## SELECTED PROJECTS

**PolyGains (Co-Developer) — polygains.com**

- Built backend and data streaming pipelines to monitor prediction market order flow and wallets.
- Engineered transaction ingestion and wallet clustering; won the Brandeis Ain Family Start-Up Award.

## EDUCATION & CREDENTIALS

MS, Business Analytics, Brandeis University Dec 2026

- Graduate researcher under Prof. Shubhranshu Shekhar, developing a dynamic profanity benchmark to overcome sociolinguistics and vernacular for adolescents age group, working on publishing a paper for NeurIPS.
- Teaching Assistant for BUS 244A 'Generative AI Foundations for Business'.
- Developed a multi-asset meta-labeling portfolio pipeline for State Street Global Advisors with Xuesong Hu, Summer 2026.

BS, Computer Science, The Open University of Israel 2022

## LEADERSHIP & SERVICE

Mentored at-risk youth through sailing and scholars through Watoobi; served 3 years as a military instructor (2014–2017).